Mini-project 5.3

Anomalous activity detection of a ship’s engine

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Date: 07-Jan-2026

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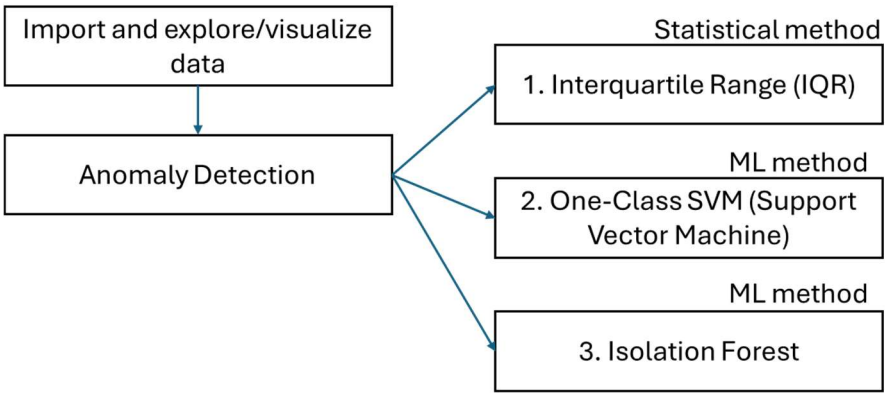
1.0 Overview: Problem Statement

The project requires detection of anomalous activity (1-5%) from a ship's engine dataset (Devabrat, 2022). Detecting this activities early and evaluating engine functionality can lead to early maintenance (eg prevention of breakdowns), thereby increasing business profits (Data Science, 2025).

2.0 Approach

The data provided was imported and exploratory data analysis (EDA) was carried out, including feature visualisation. Three different techniques were used for anomaly detection (see below). The results were analysed and visualised for accurate anomaly detection.



The data imported showed 6 features with 19,535 rows of data (snippet below).

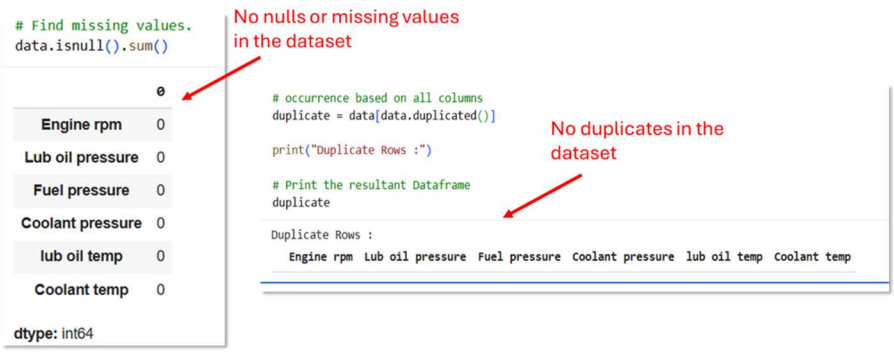
```
(19535, 6)
```

	Engine rpm	Lub oil pressure	Fuel pressure	Coolant pressure	lub oil temp	Coolant temp
0	682	2.391656	4.617196	2.848982	76.272417	69.884609
1	605	5.466877	6.424361	5.727520	73.222679	74.907314
2	658	3.434232	3.680896	1.678708	88.089916	78.704806
3	749	2.094656	7.120927	1.639670	77.661625	82.386700
4	676	3.538228	5.956472	3.225336	75.226352	67.153220

Note - *combination* of anomalies indicates potential engine issue.

2.1 Exploratory Data Analysis

Exploring the data shows there are no missing or duplicate values.



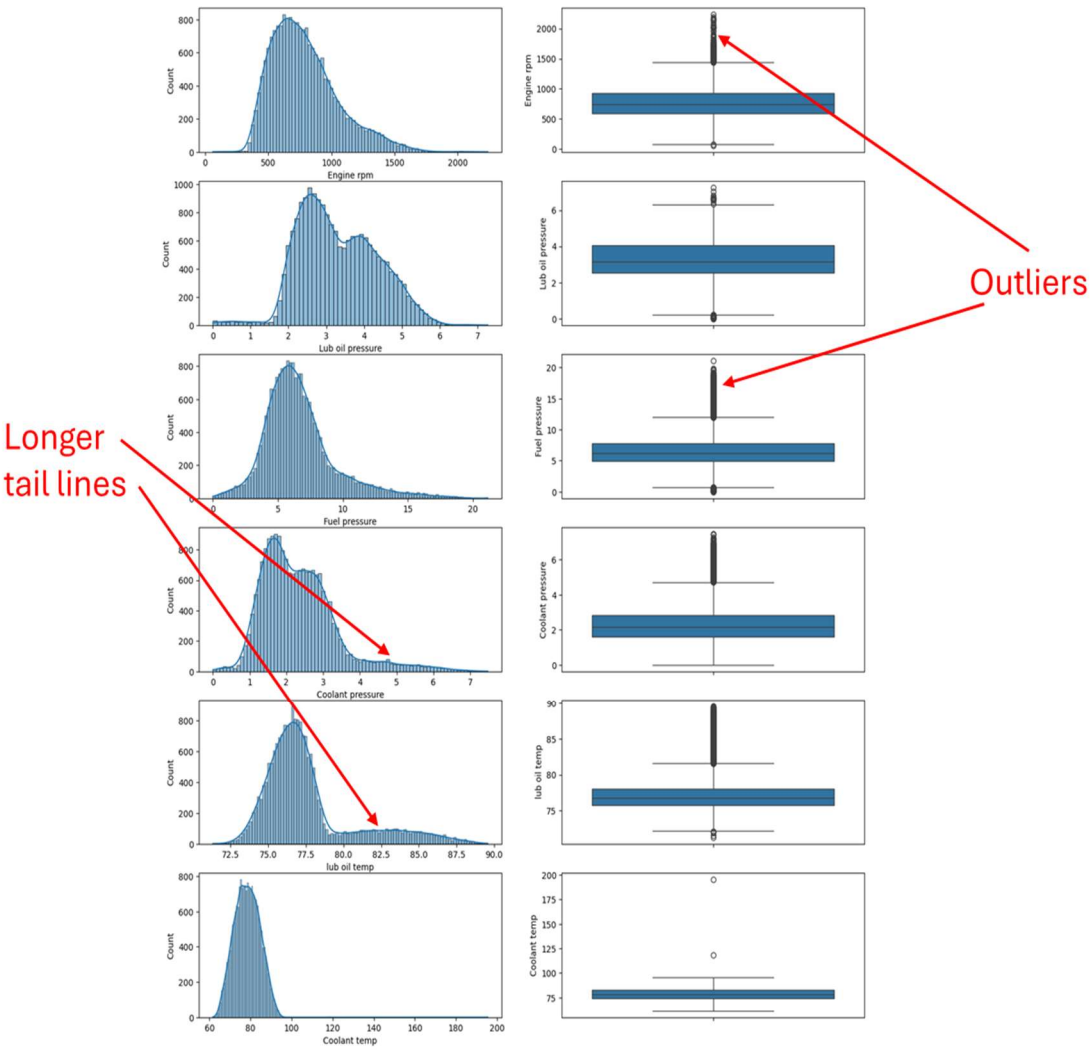
Descriptive satistics view obtained showing mean and median.

```
data.describe()
```

	Engine rpm	Lub oil pressure	Fuel pressure	Coolant pressure	lub oil temp	Coolant temp
count	19535.000000	19535.000000	19535.000000	19535.000000	19535.000000	19535.000000
mean	791.239263	3.303775	6.655615	2.335369	77.643420	78.427433
std	267.611193	1.021643	2.761021	1.036382	3.110984	6.206749
min	61.000000	0.003384	0.003187	0.002483	71.321974	61.673325
25%	593.000000	2.518815	4.916886	1.600466	75.725990	73.895421
50%	746.000000	3.162035	6.201720	2.166883	76.817350	78.346662
75%	934.000000	4.055272	7.744973	2.848840	78.071691	82.915411
max	2239.000000	7.265566	21.138326	7.478505	89.580796	195.527912

The mean (791.24) is higher for 'Engine rpm' feature, with a wide range between 61 and 2,239. The last two features means (77.64 and 78.42) are slightly higher compared to the other 3 features which are less than 7. The median (50% row) looks skewed for eg 'Coolant temp' , median is 78.35, where the minimum is 61.67 and maximum is 195.53.

Plotting each of these features in a box plot and histogram view – shows the distribution. Note the outliers, the extended tail lines of the graphs and variations. Fewer outliers on 'coolant temp', 'lub oil pressure' compared to the others. Most features have a curve like view except for 'lub oil pressure' and 'coolant pressure'. (see below)



2.2 Technique 1: IQR

The statistical method of IQR was calculated for each feature (Data Science, 2025). Data between 25th and 75th percentile is considered normal (0), else an outlier (1). Six new columns was created as per below.

	Engine rpm	Lub oil pressure	Fuel pressure	Coolant pressure	lub oil temp	Coolant temp	Engine rpm IQR	Lub oil pressure IQR	Fuel pressure IQR	Coolant pressure IQR	lub oil temp IQR	Coolant temp IQR
0	682	2.391656	4.617196	2.848982	76.272417	69.884609	0	0	0	0	0	0
1	605	5.466877	6.424361	5.727520	73.222679	74.907314	0	0	0	1	0	0
2	658	3.434232	3.680896	1.678708	88.089916	78.704806	0	0	0	0	1	0
3	749	2.094656	7.120927	1.639670	77.661625	82.386700	0	0	0	0	0	0
4	676	3.538228	5.956472	3.225336	75.226352	67.153220	0	0	0	0	0	0

Total number of combination of outliers were calculated. Results below:

Combination of features outliers	**Anomalies count**	**Out of 19, 535 data points. Require 1% to 5%: 195.35 - 976.75**
With at least 1 features as outliers	4214	22%
With at least 2 features as outliers	411	2% (falls within 1 to 5% range)
With at least 3 features as outliers	11	0.06%
With at least 4 features as outliers	0	
With at least 5 features as outliers	0	
With at least 6 features as outliers	0	

Anomalies with at least 2 features outliers listed below (411 lines). You can see which features are outliers (marked as 1 on the IQR columns).

```
data[data['Total features'] == 2]
```

	Engine rpm	Lub oil pressure	Fuel pressure	Coolant pressure	lub oil temp	Coolant temp	Engine rpm IQR	Lub oil pressure IQR	Fuel pressure IQR	Coolant pressure IQR	lub oil temp IQR	Coolant temp IQR	Total features
113	1495	3.276230	3.714396	2.495264	85.053243	75.743004	1	0	0	0	1	0	2
122	1454	2.063374	6.168260	1.260121	83.372256	82.718247	1	0	0	0	1	0	2
131	830	3.231908	13.479172	2.568096	87.480362	73.428213	0	0	1	0	1	0	2
144	1762	1.697487	4.327845	2.523372	86.750349	82.393211	1	0	0	0	1	0	2
148	1494	2.443220	3.548139	1.953403	82.234541	83.953914	1	0	0	0	1	0	2
...	...	...	...	...	...	...	...	...	...	...	...	...	...
19236	771	1.816642	12.071438	2.303487	84.841115	85.694634	0	0	1	0	1	0	2
19306	944	4.198864	5.088806	4.759353	84.348972	86.904694	0	0	0	1	1	0	2
19328	771	3.916701	5.136610	5.592643	85.757119	69.278462	0	0	0	1	1	0	2
19343	1475	2.184002	6.558549	1.265442	84.933480	88.088609	1	0	0	0	1	0	2
19433	986	2.929212	14.431486	4.985264	75.477519	69.808798	0	0	1	1	0	0	2

411 rows × 13 columns

2.3 Technique 2: One-Class SVM

Scaling is required for machine learning (ML) model One-Class SVM. This will help preventing bigger features taking priority. The dataset was scaled using standardization and new columns with the scaled data created

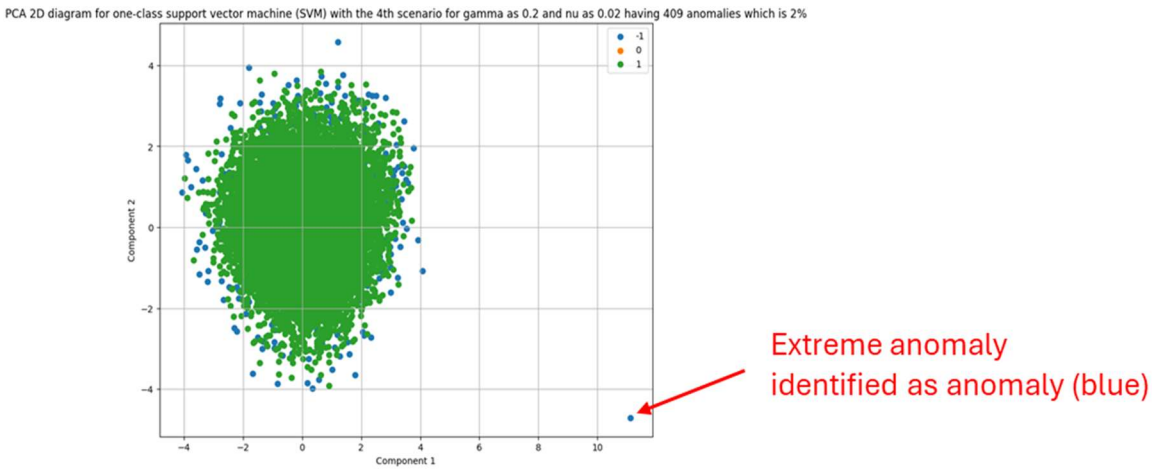
	Engine rpm SC	Lub oil pressure SC	Fuel pressure SC	Coolant pressure SC	lub oil temp SC	Coolant temp SC
0	-0.408212	-0.892819	-0.738304	0.495596	-0.440709	-1.376412
1	-0.695950	2.117332	-0.083759	3.273154	-1.421047	-0.567158
2	-0.497896	0.127697	-1.077426	-0.633625	3.358026	0.044690
3	-0.157842	-1.183534	0.168533	-0.671293	0.005852	0.637913
4	-0.430633	0.229492	-0.253226	0.858747	-0.776966	-1.816491

The scaled data was used as input into the one-class SVM model. A column was added to indicate if normal (1) or anomaly (-1). The number of anomalies were then totalled. This was repeated for different variations of the parameters used for One-Class SVM to find a variation that falls within the 1-5% anomalies range.

Gamma (influence of individual training / decision boundary)	Nu (model sensitivity to outliers)	# of anomalies (% of total dataset has to be 1-5%)
0.5	0.05	1043 (5.3%)
0.2	0.05	980 (5.01%)
0.2	0.03	594 (3.04%)
0.2	0.02	409 (2.09%)

As 2% anomalies was in IQR, the similar comparison is taken here (last row). The overlap between the 411 (2%) anomalies identified in IQR and 409 anomalies (2.09%) from one-class SVM is 116. Which is about 28%. This mean, even though they are both calling out ~2% anomalies, only 28% of those are overlapping between the two methods.

To visualize the data, Principal Component Analysis (PCA) was used helping to reduce the dimensions to 2 (Data Science, 2025).



The outliers can be seen as blue and green as the normal data points (ignore the orange legend as no dataset is marked as 0). Note some of the extreme anomalies are identified correctly by the model as blue.

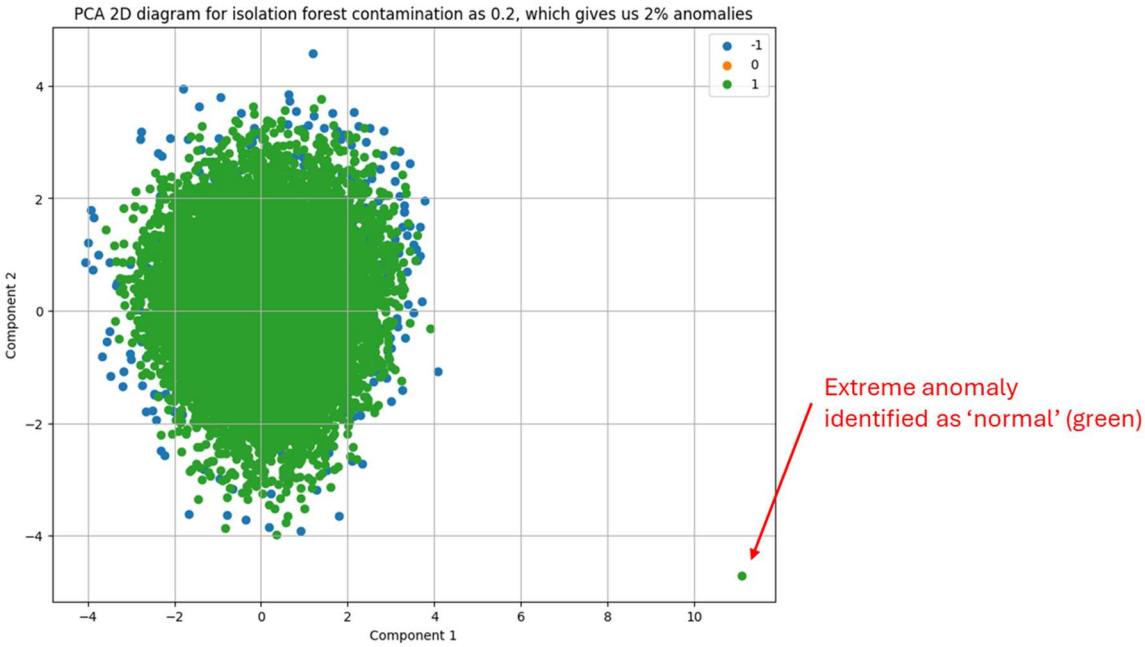
Using this PCA visualisation, you lose interpretability. i.e. its hard to trace back what the extreme anomaly links back to.

2.4 Technique 3: Isolation forest

Similarly, another ML model isolation forest was used to identify anomalies. The original 6 features dataset was used, as scaled data isn't required here. In the same way, -1 for anomalies and 1 for normal data points was captured. Contamination parameter was set to 0.02 (2% anomalies) at the start.

N_estimators (#of trees – higher can increase performance but can diminish at a certain point)	Contamination (expected ration of anomalies)	# of anomalies (needs to be 1-5%)
100	0.02	391 (2%)

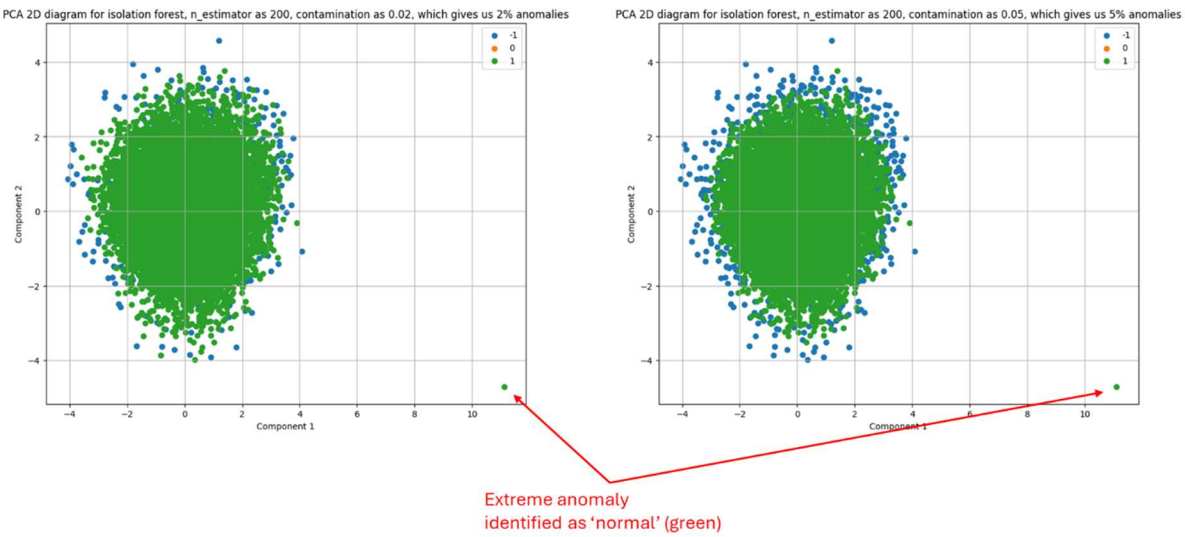
PCA was used (with the scaled data) again to visualise the results. However this time – the extreme anomaly outside the cluster comes up as a 'normal' data point. Visually, this doesn't look accurate.



Different parameters were tested to see if the green anomaly gets changed to blue in any of these:

N_estimators (#of trees – higher can increase performance but can diminish at a certain point)	Contamination (expected ration of anomalies)	# of anomalies (needs to be 1-5%)
100	0.02	391 (2%)
200	0.02	391 (2%)
200	0.05	977 (5%)

The results of the extreme datapoint on the bottom right remained the same. Bringing into question on whether if this is really accurate as visually, it looks to be an anomaly.



3.0 Conclusion and recommendations

Three different techniques were used to identify anomalies for the ships's engine. Visualisation helps to see how the data points are spread out spatially and understand the data better. PCA was used for 2D plots with positives (visually can view potential anomalies and if it is being identified correctly as an anomaly or not) and negatives (overlapping data points in clusters are hard to distinguish, and interpretability is lost).

Both the IQR and the one-class SVM are recommendations to the ship's engineer to consider using. If interpretability is required - IQR would be recommended as its easier to indicate which specific features in the data points are outliers as we didn't lose interpretability – this would be beneficial to the ship's engineer.

Next steps could be to consider any additional data that can be provided to highlight anomalies accurately and to test if these are indeed anomalies and in the real world – i.e. does it really indicate problems with the engine. Consider any other business context that could help the models that the ship's engineer could provide. Finally – the coding of the project could be further made efficient (eg IQR features could have been a function and visualised).

References

Devabrat, M. 2022. *Predictive Maintenance on Ship's Main Engine using AI.* Available at: https://dx.doi.org/10.21227/g3za-v415Links to an external site. [Accessed: 5 March 2024] *(from Cambridge data science accelerator course material)*

Data Science, 2025. *Data Science with Machine Learning & AI Career Accelerator* programme at the University of Cambridge [Accessed: 7 January 2026]